MILITARY SPECIFICATION

Cartridge, 165MM, TP, M623
Loading, Assembling and Packing

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 165MM, TP, M623.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9219045 - Cartridge, 165MM, TP, M623
8796482 - Box Packing, Ammunition, for Cartridge, 165MM, TP, M623, HEP, M123A1 in fiber Container M387
8796483 - Container Fiber, Ammunition, M387, for Cartridge, 165MM, TP, M623, HEP, M123A1

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with the applicable drawings and specifications.

3.2 Complete round.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 9219045, and with all requirements specified in the applicable specifications.

3.3 Propellant weight.-The weight of the propellant shall not vary by more than 0.10 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.4 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.5.5.

3.4.1 Muzzle velocity.-The muzzle velocity of the cartridge when fired at seventy degrees Fahrenheit (70°F)(21.1 degrees Celsius (°C)) shall not be more than 875 feet per second (fps) (266.7 meters per second (mps)), nor less than 825 fps (251.5 mps), and the lot velocity standard deviation shall not be greater than 10.2 fps (3.1 mps).

3.4.2 Safety.-There shall be no hangfires, or indication of metal parts failure in the gun bore or in flight, prior to impact, during any of the proving ground tests conducted.

3.5 Leakage testing.-The heat-sealed propelling charge assembly shall be air leak tested in a vacuum pressure differential chamber. Each assembly must not exceed a leakage rate of 0.05 cubic centimeter per second.

3.6 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

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3.7.1 General.-The cartridge and all components shall be free of cracks or burrs; and chips, dirt, grease, rust or other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The cartridge shall not be damaged in any manner during the loading operation.

3.7.2 Painting and marking.-The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated. (Bare spots in painting, other than marking, are allowable provided that they total not more than 1/4 square inch in aggregate, and that the maximum dimension of any individual bare spot does not exceed 3/16 inch.)

3.7.3 Projectile loading.-The interior surface of the projectile body shall be clean and dry when the inert material is loaded into it. The threads shall be free of inert material and foreign material. If a chemical cleaner is required, no solvent other than acetone shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this item detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Cartridge Assembly	9219045	55
Propelling Charge Assembly	8845054	25
Liner	8845062	25
Liner Cover	8845063	25
Suppl. Ignition Charge	8845064	25
Bag	8845065	25
Washer	9234094	25

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4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, 165MM, TP, M623				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Washer (Dwg. 9234094)				
	Examination for defects	25			4.4.2.1
	Bag (Dwg. 8845065)				
	Examination for defects	25			4.4.2.2
	Supplementary Ignition Charge (Dwg. 8845064)				
	Examination for defects	25			4.4.2.3
	Cover, Liner (Dwg. 8845063)				
	Examination for defects	25			4.4.2.4
	Liner (Dwg. 8845062)				
	Examination for defects	25			4.4.2.5
	Propelling Charge Assembly (Dwg. 8845054)				
	Examination for defects	25			4.4.2.6, 4.4.2.7
	Leakage Test	25			

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	CONTINUATION SHEET				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 165MM, TP, M623, Assembly (Dwg. 9219045)				
	Examination for defects	25			4.4.2.8, 4.4.2.9, 4.4.2.10, 4.4.2.11, 4.4.2.12, 4.4.2.13.
	Electrical continuity (Note 1)	25			4.5.3
	Primer seating (Note 1)	25			4.5.4
	Ballistic test				
	Muzzle velocity	10			4.5.5.1, 4.5.5.3
	Security				
	125° - 5°F	10			4.5.5.2, 4.5.5.3
	-40° + 5°F	10			4.5.5.2, 4.5.5.3

NOTES:

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation.-Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of cartridges shall contain:

a. Cartridge cases of one lot interfix number from one manufacturer;
b. Projectile metal parts from one lot interfix number from one manufacturer;
c. Propelling charges assembled from one lot of propellant from one manufacturer;
d. Loaded primers of one lot interfix number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Washer		9234094
			NEXT HIGHER ASSEMBLY 9219046

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter		0.65%	3.1	Gage
202	Thickness		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Bag		8845065
			NEXT HIGHER ASSEMBLY
			8845064

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
Minor					
201	Length of cross seam thread projection		0.65%	3.1	Gage
202	Distance of cross seam from edge of bag		0.65%	3.1	Gage
203	Distance of side seam from edge of bag		0.65%	3.1	Gage
204	Stitch missing or incorrect		0.65%	3.1	Visual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Charge, Ignition, Supplementary, For Cartridge, 165MM, HEP	1 OF 1	8845064
			NEXT HIGHER ASSEMBLY 8845054

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charge will not freely assemble over mandrel		0.40%	3.2	Manual
102	Any opening or tear in bag large enough to allow ignition charge to escape		100%	3.2	Visual
Minor					
201	Charge weight incorrect		0.65%	3.1	Balance
202	Length of cross seam thread projection		0.65%	3.1	Gage
203	Distance of cross seam from edge of bag		0.65%	3.1	Gage
204	Stitch missing or incorrect		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Cover, Liner (See Note)		8845063
			NEXT HIGHER ASSEMBLY 8845054

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of shell rib		0.40%	3.1	Gage
102	Outside diameter of flange		0.40%	3.1	Gage
103	Thickness of flange		0.40%	3.1	Gage
Minor					
201	Large inside diameter of recess		0.65%	3.1	Gage
202	Small inside diameter of recess		0.65%	3.1	Gage
203	Inside diameter of center recess		0.65%	3.1	Gage
204	Distance from flange to back of recess		0.65%	3.1	Gage
205	Distance from top of rib to back of flange		0.65%	3.1	Gage
206	Distance from top of center recess to back of flange		0.65%	3.1	Gage
207	Thickness of rib		0.65%	3.1	Gage
208	Body thickness		0.65%	3.1	Gage
209	Radii missing or incorrect		0.65%	3.1	Visual
210	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES: 1. See 4.4.3.1 For optional sampling plan.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Liner (See Note)		8845062
			NEXT HIGHER ASSEMBLY 8845054

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.1	Gage
102	True position of small cavity with outside diameter		0.40%	3.1	Gage
103	True position of large cavity with outside diameter		0.40%	3.1	Gage
Minor					
201	Diameter of small cavity		0.65%	3.1	Gage
202	Diameter of large cavity		0.65%	3.1	Gage
203	Thickness of body		0.65%	3.1	Gage
204	Thickness of small cavity sidewall		0.65%	3.1	Gage
205	Thickness of small cavity bottom		0.65%	3.1	Gage
206	Length, total		0.65%	3.1	Gage
207	Distance to large cavity shoulder from rear of body		0.65%	3.1	Gage
208	Distance to bottom of small cavity from rear of body		0.65%	3.1	Gage
209	Profile of rear of body		0.65%	3.1	Gage
210	Radii missing or incorrect		0.65%	3.1	Visual
211	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES: 1. See 4.4.3.1 for optional sampling plan.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Propelling Charge Assembly, Prior to Sealing		8845054
			NEXT HIGHER ASSEMBLY 8845042

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant charge weight		100%	3.3	Balance
102	Igniter charge missing		100%	3.2	Visual
103	Presence of foreign matter		0.40%	3.7.1	Visual
104	Evidence of propellant powder grains between center of liner and cover		0.40%	3.2	Visual
Minor					
201	Evidence or poor workmanship		0.65%	3.7.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Propelling Charge Assembly, After Sealing				8845054
					NEXT HIGHER ASSEMBLY
					8845042
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly not sealed full 360°		100%	3.1	Visual/Manual
102	Leakage test		100%	3.5	Test/4.5.1
Minor					
201	Outside diameter, max.		0.65%	3.1	Gage
202	Length, total, max.		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Projectile Loading Assembly, Prior to Assembly of Base Plug	1 OF 1	9219046
			NEXT HIGHER ASSEMBLY: 9219045

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Filler missing or improperly poured		0.40%	3.2	Visual
102	Depth to top of filler		0.40%	3.2	Gage
Minor					
201	Foreign matter, including inert filler, on threads		0.65%	3.7.3	Visual
202	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Projectile Loading Assembly		9219046
			NEXT HIGHER ASSEMBLY 9219045

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Base plug fails to withstand reapplication of minimum assembly torque (while adhesive is wet)			3.2	Gage
Major					
101	Projectile body diameter, max.		100%	3.2	Gage
102	Adhesive missing from base plug threads prior to assembly of projectile		100%	3.2	Visual
103	Projectile body ogive contour improper		0.40%	3.2	Gage
Minor					
201	Filler on exterior of projectile body		0.65%	3.7.3	Visual
202	Marking incorrect, missing or illegible		0.65%	3.7.2	Visual
203	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Cartridge Case, After Assembly of Primer, Primer Spring and Spring Retainer	1 OF 1	9219045
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Electrical continuity		100%	3.2	Test/4.5.2
102	Primer seating		100%	3.2	Test/4.5.4
103	Spring retainer not crimped for full 360°		0.40%	3.2	Visual
104	Any component missing or improperly assembled		0.40%	3.2	Visual
105	Primer check test	a		3.1	
Minor					
201	Foreign matter on interior of cartridge case		0.65%	3.7.1	Visual
202	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:
a. See 4.4.3.2 for sample size and test procedure.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Cartridge Case Assembly, After Insertion of Propelling Charge, Prior to Assembly to Projectile	1 OF 1	9219045
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner damaged, exposing propelling charge or permitting charge to escape			3.2	Visual
Major					
101	Foreign matter on body threads		0.40%	3.7.3	Visual
102	Propelling charge assembly not properly seated		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Assembly, Prior to Assembly of Handle		9219045
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Electrical continuity		100%	3.2	Test/4.5.3
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Cartridge, 165MM, TP, M623 Assembly		9219045
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component loose		0.40%	3.2	Manual
102	One or more stakes missing		0.40%	3.2	Visual
103	Gap between cartridge case shoulder and projectile body shoulder extending more than half of the circumference		0.40%	3.2	Visual
Minor					
201	Total length, max. (with handle folded)		0.65%	3.2	Gage
202	Handle assembly improperly seated		0.65%	3.2	Visual
203	Rotating band nicked, gouged or otherwise damaged		0.65%	3.2	Visual
204	Projectile body protective coating with bare spots		0.65%	3.7.2	Visual
205	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Unsealed Fiber Container		8796483
			NEXT HIGHER ASSEMBLY 8796482

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Nose or rear support missing		0.40%	3.1	Visual
Minor					
201	Cartridge cannot be removed by hand		0.65%	3.1	Manual
202	Pad or filler missing		0.65%	3.1	Visual
203	Glue or asphalt on cartridge		0.65%	3.1	Visual
204	Container cap cannot be removed by hand		0.65%	3.1	Manual
205	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Sealed Fiber Container		8796483
			NEXT HIGHER ASSEMBLY 8796482

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Sealing tape incomplete		0.40%	3.1	Visual
102	Metal end loose or damaged		0.40%	3.1	Visual/Manual
103	Container cut or damaged through all asphalt impregnated layers		0.40%	3.1	Visual/Manual
Minor					
201	Sealing tape tab length, min.		0.65%	3.1	Gage
202	Contents loose		0.65%	3.1	Manual
203	Gap between cover and body of container exceeding one-eight inch		0.65%	3.1	Visual
204	Cuts, scuffs or gouges in outer layers		0.65%	3.1	Visual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50413A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.16	Sealed Wood Packing Box		8796482
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged, exposing contents		0.40%	3.1	Visual
102	Hardware or strapping missing, broken or loose		0.40%	3.1	Visual/Manual
103	DoD symbol missing, incorrect or illegible		0.40%	3.1	Visual
Minor					
201	Hardware or strapping improperly assembled		0.65%	3.1	Visual
202	Handle missing or loose		0.65%	3.1	Visual/Manual
203	Car seal missing, unsealed or mispositioned		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.7.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

4.4.3.1 Dimensional control of molded parts.-In place of the normal sampling associated with the classification of defects, and subject to the approval of the contracting officer, the optional sampling plan in paragraph 4.4.3.1.1 may be used for dimensional control of molded parts. Use of this optional plan is limited to those dimensions and features controlled by the mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the mold cavity, these optional plans are not applicable. In the event of in-process rejects during optional sampling, corrective action applicable to parts produced prior to those rejected, and subsequent to the last acceptable sample, shall require approval of the Contracting Officer.

4.4.3.1.1 Sampling plan.-A sample of five parts (as molded) shall be inspected to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part will be removed from production. Further, that portion of production since the last control check shall be rejected, pending corrective action. All cavities removed from production because of some fault, may, after reworking, be returned to production, providing they pass the qualification test above.

4.4.3.2 Check test for possible deterioration of primers, Major defect.-If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable primer specification immediately before the primer lot is assembled into the cartridges (see 6.6). Examples of exposure of primers to adverse conditions are indications of corrosion products on the primer body; moisture (water) marks or stains on the exterior packaging; or, cuts, scrapes, scuff marks, gouges or tears in the inner packaging, including the barrier bag material's integrity and security of the seal.

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4.4.3.3 Proving ground.-The following tests shall be performed as specified in 4.5.5. If any sample fails to meet the criteria for safety as prescribed in 3.4, the lot represented by that sample shall be rejected.

4.4.3.3.1 Muzzle velocity, major defect

4.4.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, the sampling plan and acceptance for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I and an AQL of 1.00 percent. In addition, if the standard deviation of the sample's velocity multiplied by the factor 0.75 exceeds 10.2 fps, the lot shall be rejected.

4.4.3.3.1.2 After three consecutive lots have met the criteria of 4.4.3.3.1.1, the sampling plan and acceptance for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter F, and an AQL of 1.00 percent. In addition, if the standard deviation of the sample's velocity multiplied by the factor 0.65 exceeds 10.2 fps, the lot shall be rejected.

4.4.3.3.2 Security, critical defect.-Twenty cartridges shall be selected from each lot and fired for security as specified in 4.5.5.2. Ten samples shall be fired at 125 degrees Fahrenheit and minus 40 degrees Fahrenheit.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.2 herein and Section 6 of MIL-A-48078).

4.5 Test methods and procedures

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Leakage test.-Each propelling charge assembly shall be tested for leakage using a partial vacuum system utilizing a chamber with minimal clearance between the exterior of the liner and interior of the chamber. Water immersion testing shall not be performed, as such a test shall be considered as being destructive to any charge assembly so tested. The contractor shall prepare a

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detailed written standard operating procedure and description of equipment, including test and calibration procedures, which must be submitted to and approved by the Contracting Officer prior to the submission of the first article sample.

4.5.2 Electrical continuity, prior to insertion of propelling charge.-The primed cartridge case shall be tested for electrical resistance by applying an electrical current of twenty milliamperes, max., using the igniter button as one contact point and the projectile body thread as the other contact point.

4.5.3 Electrical continuity.-The complete cartridge, less handle assembly, shall be tested for electrical resistance by applying an electrical current of twenty milliamperes, max., using the igniter button as one contact point and a position one-half inch forward of the rotating band on the projectile body as the other contact point.

4.5.4 Primer seating.-The primer shall be manually depressed against primer spring tension to its furthest extent and upon release shall freely return so that the forward wall of the circumferential groove is flush or above flush with the adjacent cartridge case surface.

4.5.5 Proving ground tests.-The samples which are to be ballistically tested as prescribed herein shall be fired in a M135 gun. The ammunition in each phase shall be conditioned to the appropriate temperature until all propellant is stabilized at a uniform temperature prior to firing, which normally requires a period of twenty-four hours, minimum. Photographic equipment shall be used to assure satisfactory coverage of each test sample's integrity and flight from the gun muzzle to the first one-hundred feet of projectile flight, minimum. Muzzle velocity shall be recorded for all rounds, including reference ammunition and spotters; and, maximum chamber pressure shall be recorded for all reference and security phase rounds. All samples shall be observed for safety (see 6.5).

4.5.5.1 Muzzle velocity.-The data generated in this phase will be used for the acceptance of a lot from a muzzle velocity standpoint and will require the firing of reference ammunition to establish a velocity correction. The firing sequence shall be five reference, ten test, and five reference rounds. The ammunition fired in this phase shall be temperature conditioned to the temperature of $70^{\circ} \pm 2.5^{\circ}$F($21.1^{\circ} \pm 1.5^{\circ}$ Celsius). All rounds shall be observed for target impact dispersion on a vertical cloth target, for information only.

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4.5.5.1.1 Target impact dispersion

4.5.5.1.1.1 Target size and distance.-The vertical cloth target distance is to be 900 meters and the target size is to be 6 meters by 6 meters, minimum. No more than one lot acceptance test is to be performed on each target.

4.5.5.1.1.2 Grouping and cadence.-The test and reference cartridges shall be fired in five round groups (occassions), with a cadence of not less than three minutes and not more than five minutes between successive firings. A cadence of less than three minutes may be used; however, this cadence must be maintained for all rounds to be fired on that occassion.

4.5.5.1.1.3 Removal of backlash from fire control systems.-After loading and locking a round in the gun chamber, the gunner shall traverse the gun off the center of the target, drop down below target center, traverse to the side opposite target center, elevate to target center, and traverse the gun back onto target center. If the gun sight indicates that the gun is aimed above or beyond target center, do not reverse the direction of the movement of the gun; but, instead, repeat the above traversing sequence. The distance traversed and dropped should be sufficient to remove backlash from the fire control system.

4.5.5.1.1.4 Spotting the target.-Spotter rounds will be used to assure that the weapon is zeroed to assure engagement of target center. A minimum of three consecutive target hits by spotters which had been fired at the same elevation and deflection are required. An aiming point or reference cartridges may be used for this purpose. Once the weapon has been spotted on the target for a shot group, the elevation and deflection of the weapon thus determined shall remain the same for that shot group.

4.5.5.1.1.5 Wind velocity.-Wind velocity and gusts shall be measured at the firing point and at the 900 meter target prior to and after the velocity/target dispersion test group, and at the firing point prior to firing each round of the group. The velocity/dispersion portion of the ballistic test shall not be fired if the surface wind velocity (nominal) or surface wind gusts exceed 10 knots (18.5) km/hr and 5 knots (9.26 km/hr) respectively, at either equipment location. Only that component of the surface wind normal to the line of fire will be considered. In addition, peak surface wind velocity (nominal plus gusts) normal to the line of fire is not to exceed 10 knots (18.5 km/hr) during the firing of any shot group during the velocity/dispersion phase.

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4.5.5.1.1.6 Calculation of probable error.-Target impacts shall be measured to the nearest inch (2.5 centimeters) in both the horizontal and vertical distance from the center of the target. The vertical and horizontal hit pattern results for each shot group shall be expressed as probable errors in mils, pooled, and then recorded on the firing record (see 6.5.1 for calculation of pooled probable error).

4.5.5.2 Security.-Half of the cartridges selected for the security phase shall be temperature conditioned to 125° - 5°F (51.6° - 3.0°C) and half shall be temperature conditioned to -40° + 5°F (-40° + 3.0°C). Each round fired shall be observed for safety. Muzzle velocity shall be recorded for information.

4.5.5.3 Safety.-All cartridges fired in proving ground tests shall be observed for premature bursts, hang-fires or breakup of projectile metal parts, including separation of the cartridge case from the projectile body. Any ballistic sample failing the safety criteria herein shall be cause for the cessation of the test phase in which the failure occurs, as well as, the remainder of the ballistic test, and shall reject the lot represented by that ballistic sample.

4.5.5.4 Retest.-If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges, as required, shall be tested. A lot shall be retested whenever the proving ground deems it necessary, or whenever any of the following conditions exist:

a. The muzzle velocity correction factor computed from the average muzzle velocity of the reference rounds exceeds 25 fps;
b. The velocity standard deviation of the reference rounds, multiplied by the factor 0.65, exceeds 10.2 fps;
c. Two or more velocities are lost (test sample or reference round) when firing the muzzle velocity phase.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 8796483.

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5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 8796482.

5.3 Marking

5.3.1 Level A.-The fiber containers and wood packing boxes shall be marked and sealed in accordance with dwgs. 8796482 and 8796483.

6. NOTES

6.1 Ordering data

6.1.1 Procurement requirements.-See MIL-A-48078.

6.1.2 Delivery of data to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy, each, to the technical activity, of the Ammunition Data Cards prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.2 Submission of designs for approval.-See paragraph 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Changes from previous issues.-Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

6.4 Visual examinations.-When compliance with the applicable requirements is in doubt, as a result of a visual examination, the affected characteristic may be measured or gaged to determine acceptability.

6.5 Firing record information.-The ballistic test firing records shall include the serial numbers of all weapon and fire control components. Stock component lot numbers of warmers and spotters shall also be included. For each round fired, the record shall contain deflection, elevation, time of firing, chamber pressure (where required), target impact coordinates, pertinent flight characteristics and meteorological data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E and DRDAR-LCU-CV, Dover, New Jersey 07801.

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6.5.1 Method for calculating pooled probable error.-The individual probable error (PE) results of each of the "n" groups fired shall be pooled according to the following method:

$$\text{Pooled PE} = \left[\frac{\sum_{i=1}^{n} (SS_i - 1)(PE_i)^2}{\sum_{i=1}^{n} (SSi - 1)} \right]^{1/2}$$

Where PE_i and SS_i are the probable error value and sample size, respectively, for each of the "n" groups of five samples each which were fired for muzzle velocity/target impact dispersion data.

6.6 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge and shall not constitute a basis for rejection against either contractor, except where deteioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Proving ground test summary

Phase	Sample Size First Article	Sample Size Production	Requirement	Sampling Paragraph	Test Procedure Paragraph
Velocity (70°F)	10	10*	Muzzle Velocity	4.4.3.3.1	4.5.5.1
			Vel. Std. Dev.	4.4.3.3.1	4.5.5.1
			Safety	4.4.3.3.2	4.5.5.3
			TID (Info)		4.5.5.1.1
Security					4.5.5.2
-40°F	10	10	Safety	4.4.3.3.2	4.5.5.3
+125°F	10	10	M.V.(Info)		

*The sample size for the first three consecutive acceptable lots shall be 25 cartridges.

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Preparing Activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-50413A Cartridge, 165MM, TP, M623 LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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